

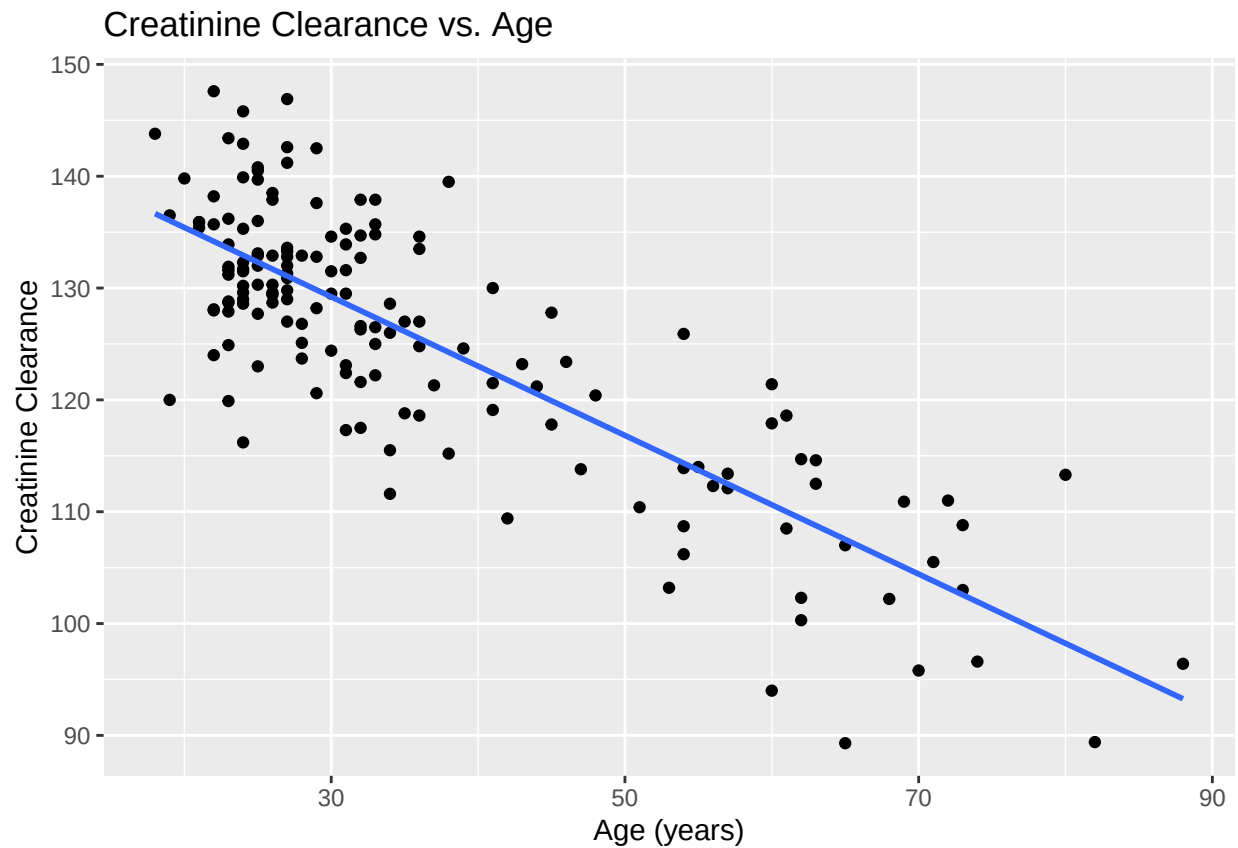
Homework 8

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2025-04-14

GitHub Link:

Problem 1: Regression Warm up



Part A

The expected creatinine clearance rate for a 55-year-old is approximately **113.71 mL/minute**. This expected rate was determined by obtaining the regression equation below from the visualization of the linear regression line above.

Equation:

- $\hat{y} = 147.81 - 0.62x$

Part B

For every year of age, creatinine clearance **decreases** by approximately **-0.62 mL/minute**.

- I obtained this value by fitting a linear regression model using the `lm()` function, using creatinine clearance as the response variable and age as the explanatory variable. Then, I utilized the `coef()` function to obtain the slope of the regression line.

Part C

Based on the regression model ($\hat{y} = 147.81 - 0.62x$), the predicted creatinine clearance rate of a 40-year old is 116.82 mL/minute, and the predicted creatinine clearance rate for a 60-year old is 110.62 mL/minute. Comparing the actual values with the predicted, we see:

- 40-year-old: $135 - 116.82 = 18.18$ mL/min above expected
- 60-year-old: $112 - 110.62 = 1.38$ mL/min above expected

The creatinine clearance rate is healthier in the 40-year-old with a rate of 135 compared to the 60-year-old.

Problem 2: Modeling Disease Growth

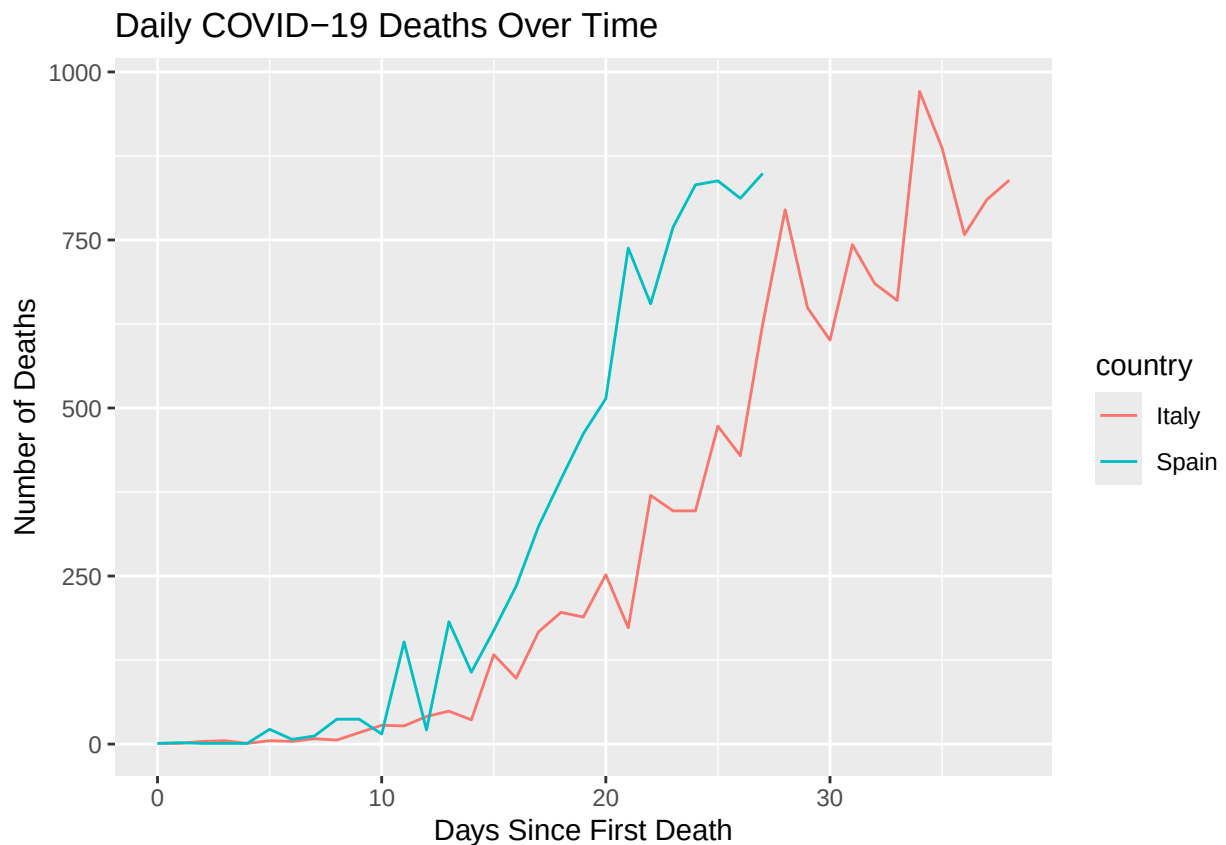
Part 1

The estimated growth rate and doubling time for Italy is 0.183 and 3.8 days, respectively. The 95% bootstrapped confidence intervals for both measurements are (0.159, 0.209) and (3.323, 4.367), respectively.

Part 2

The estimated growth rate and doubling time for Spain is 0.276 and 2.5 days, respectively. The 95% bootstrapped confidence intervals for both measurements are (0.234, 0.320) and (2.168, 2.958), respectively.

Part 3



Problem 3: Price Elasticity of Demand

The estimated price elasticity of demand for milk is approximately **-1.619**, meaning that the demand for milk decreases as the price increases. The estimated 95% bootstrapped confidence interval is **(-1.784, -1.460)**.

To estimate the price elasticity of demand for milk, I fit a linear regression model using $\log(\text{sales})$ and $\log(\text{prices})$ based on the power-model equation, $Q_t = K * P_t^{\beta}$, from which I obtained the price elasticity estimate. Then, I used the bootstrapping technique to get a 95% confidence interval for the price elasticity.